Reading HP8595E Calibration Constants

Tom DG8SAQ, 24.1.2025

The GPIB command *CAL DUMP* allows to access analyzer calibration constants in unformatted numeric form. The trick is to identify which number belongs to which cal constant.

Also useful is the chapter "*Backing Up Analyzer Correction* Constants" on pages 201ff in the *8590 Series Analyzers Assembly-Level Repair* manual 08590-90316.

Helpful for identifying the CAL DUMP data is the analyzer screen obtained via menu DISPLAY CAL DATA.

Here are screenshots of Wilko's HP8595E cal constants as of 19.1.12025:

Page 1

```
CMD ERR:IDN
                                                                  NEXT
       BANDWIDTH           AMPLITUDE                              PAGE

  6dB  BWAMP LC XTL  RFATN  SGAIN  LOG   LIN   NBW
200H -0.04    0 132   0.08    0dB -0.01                           DACS
  9K  0.80 128 128   0.00   10dB  0.00  0.51  0.00
120K  0.53  41 255   0.06   20dB  0.44  0.36  0.00
                     0.01   30dB  0.46  0.61  0.00             STP GAIN
  3dB  BWAMP LC XTL  0.13   40dB  0.66  0.55  0.00                 ZERO
 10H  1.52 128 128   0.06   50dB  0.67
 30H  1.28 128 128   0.12   60dB
100H  0.88 128 128   0.36   70dB   MCDLY       140            ANALYZER
300H  0.42    0 255                                              GAINS
  1K  0.11    0 223    CA ATT ERR    BND GAIN ERR
  3K  0.00    0 165      1 -0.08      0 176 -0.09
 10K -0.12    0 100      2 -0.06      1 176 -0.09
 30K -0.42    0  27      4  0.09      2 176 -0.09
100K  0.69   53 255      8 -0.01      3 176 -0.09
300K  0.68  109 255     16  0.00      4 176 -0.09
  1M  0.49  170 255                                              More 1
  3M  0.42  231 255  HRM      1
 ~5M  0.41  242 255  MXRB  1760
                                                                   RT
```

Page 2

```
CMD ERR:IDN IDN
                                                               DISPLAY
                  TUNING                                      CAL DATA

  CAL     300000000  Sweepsens <10M 0.000173758
  ZERO    132581632  Sweepsens Wide 0.000001007                  DACS
  FAST     54148252  Main Coil Sens 0.059411742
  MED    1067623152  FM Coilsens Er 1.498162508
  SLOW     21459318  Wide Disc Err  4.178965569
  PkOfst        -15  Wdsc sweepsens 0.000485602               STP GAIN
  TCXO          116  EYO Δ slope         3.604E7                  ZERO
  Last Cal Freq 20:48:34  05 DEC 2024
  Last Cal Amp  20:52:21  05 DEC 2024
TRACKING GEN      FM DEMODULATION                             ANALYZER
                                                                 GAINS
AOFST         219 ZERO           0
XOFST        3095 SLOPE      17255
DCORR      -32640 Narrow BW   1927
ASLOP 1.000000000 Wide BW     1883
FSLOP 0.000000066
XSLOP 0.307900012                                                More 1
GATE 1.000000000
                                                                   RL
```

HP8595E Cal Dump:
Numbers in order of CAL DUMP output, entries identified and denoted as in screens above (page,column).
(Required) => needed for recovery after NVRAM loss and auto-cal.

CAL (2,1)
300000000
Zero (2,1)
132581632
FAST (2,1)
54148252
SLOW (2,1)
21459318
Sweepsens <10M (2,2)
1.737578E-04
Sweepsens Wide (2,2)
1.006844E-06
MED (2,1)
1067623152
FM Coilsens Er (2,2)
1.498163
GATE (2,bott)
1
Wide Disc Err (2,2)
4.178966
Main Coilsens Er (2,2)
5.941174E-02
Wdsc Sweepsens (2,2)
4.856015E-04
PkOfst (2,1)
-15
MCDLY (1,R)?
140

Unknown (2 entries)
1
65

Tracking generator constants (2,1)
AOFST
219
DCORR
-32640
XOFST (required if TG is present)
3095
ASLOP
1
FSLOP
6.625001E-08
XSLOP (required if TG is present)
0.3079

Unknown (5 entries)
200
200
200
200
200

BND Gain (1,5)

176
176
176
176
176

Unknown (5 entries)
130
130
130
130
130

TCXO (2,1) unsigned byte (required if no OCXO)
116

CA ATT ERR (1,4)
-8E-2
-6E-2
9E-2
-1E-2
0

BND GAIN ERR (1,6)
-9E-2
-9E-2
-9E-2
-9E-2
-9E-2

Unknown (6 entries)
0
0
0
0
0
0

BWAMP (3,1) 200H, 9K
-4E-2
.80
BWAMP (3,1) 10H…5M
1.52

1.28
.88
.42
.11
0
-.12
-.42
.69
.68
.49
.42
.41
BWAMP (3,1) 120K
.53

LC (3,2) 200H, 9K
0
128
LC (3,2) 10H…5M

128
128
128
0
0
0
0
0
53
109
170
231
242
LC (3,2) 120K
41

XTL (3,3) 200H, 9K
132
128
XTL (3,3) 200H, 9K
128
128
128
255
223
165
100
27
255
255
255
255

255

XTL (3,3) 120K

255

SGAIN LOG (3,5)

-1E-2
0
.44
.46
.66
.67

SGAIN LIN (3,6)

.51
.36
.61
.55

RFATN (3,4) (required)

8E-2
0
6E-2
1E-2
.13
6E-2
.12
.36

Unknown 129 entries, looks like some level correction data

-.50
-.50
-.50
-.50
-.50
-.50
-.50
-.50
-.14
.21
.45
.69
.83
.95
1.07
1.14
1.21
1.26
1.27
1.27
1.28
1.30
1.31

1.32
1.32
1.32
1.32
1.33
1.34
1.36
1.35
1.34
1.33
1.33
1.34
1.34
1.34
1.35
1.35
1.36
1.36
1.37
1.38
1.36
1.34
1.33
1.30
1.28
1.26
1.24
1.22
1.20
1.19
1.18
1.17
1.17
1.16
1.15
1.14
1.13
1.12
1.10
1.08
1.05
1.02
1.00
.98
.96
.94
.92
.91
.90
.90
.90
.90

.89
.88
.87
.85
.83
.81
.79
.77
.75
.73
.72
.70
.70
.71
.72
.72
.71
.71
.70
.70
.69
.69
.68
.67
.66
.66
.65
.65
.64
.64
.63
.62
.60
.58
.55
.52
.49
.46
.46
.46
.46
.47
.48
.49
.49
.49
.49
.50
.52
.54
.57
.60

.64
.67

Flatness Data (required)
Flatness Band 0:
Range: Start, Stop, Increment in Hz
12000000
2892000000
72000000
Flatness Data (41 entries)
-.79
-.23
-.23
-.11
-2E-2
.12
.40
.38
.94
1.00
1.59
1.21
1.59
1.33
1.59
1.56
1.42
1.52
1.80
2.19
1.87
2.10
1.91
2.52
2.10
3.01
2.42
3.33
3.01
4.28
3.87
4.40
4.10
4.26
3.84
3.66
3.42
3.75
3.63
4.01
4.80

Flatness Band 1:
Range: Start, Stop, Increment in Hz
2750000000
6508400000
234900000
Flatness Data (17 entries)
2.81
2.26
1.85
1.45
1.92
1.82
2.08
1.96
1.78
1.59
1.45
1.28
1.42
1.47
1.61
1.71
1.69
End of Flatness indicator
0
HRM MXRB (3)
1760

END